Sri Lanka Institute of Information Technology



IT3091 – MACHINE LEARNING

Dataset & Data Dictionary

Group Number: 2026-DS-35

Student ID	Student Name
IT24104087	Rathnamalala R.I.B.T
IT24103907	Pamindi M.M.
IT24400066	Athauda P.A.K.D.
IT24103888	Gunathunga K.S.N.T.

Dataset overview

The notebook uses the DataCo Supply Chain dataset with 180,519 records and 53 original columns. The modelling target is Late_delivery_risk (1 = late, 0 = not late). The class split is 54.83% late and 45.17% not late.

Data-quality summary

Product Description is fully missing. Order Zipcode is heavily missing. Customer Lname has 8 missing values and Customer Zipcode has 3. No fully duplicated rows were found. Leakage, PII and identifier fields are removed before modelling.

Column	Type	Working definition	Notebook treatment
Type	object	Payment transaction type	Keep; one-hot encode
Days for shipping (real)	int64	Actual shipping duration in days	Drop: post-outcome leakage
Days for shipment (scheduled)	int64	Scheduled shipping duration in days	Keep numeric
Benefit per order	float64	Profit/benefit generated by an order	Keep numeric
Sales per customer	float64	Sales value associated with the customer/order	Keep numeric
Delivery Status	object	Observed delivery outcome/status	Drop: directly reveals target
Late_delivery_risk	int64	Binary target: 1 = late, 0 = not late	Target
Category Id	int64	Category identifier	Drop: ID
Category Name	object	Product category name	Keep; smoothed target encoding
Customer City	object	Customer city	Keep; smoothed target encoding
Customer Country	object	Customer country	Keep; low-cardinality categorical
Customer Email	object	Customer email address	Drop: PII
Customer Fname	object	Customer first name	Drop: PII
Customer Id	int64	Customer identifier	Drop: ID
Customer Lname	object	Customer last name	Drop: PII
Customer Password	object	Customer password field	Drop: sensitive PII
Customer Segment	object	Customer segment/category	Keep; one-hot encode
Customer State	object	Customer state	Keep; smoothed target encoding
Customer Street	object	Customer street address	Drop: PII
Customer Zipcode	float64	Customer postal/ZIP code	Drop: PII; a few missing
Department Id	int64	Department identifier	Drop: ID
Department Name	object	Department name	Keep; one-hot encode
Latitude	float64	Geographic latitude	Keep numeric
Longitude	float64	Geographic longitude	Keep numeric
Market	object	Sales market/region group	Keep; one-hot encode
Order City	object	Order destination city	Keep; smoothed target encoding
Order Country	object	Order destination country	Keep; smoothed target encoding

Order Customer Id	int64	Customer identifier attached to order	Drop: ID
order date (DateOrders)	object	Order timestamp	Convert to datetime; derive month/day-of-week
Order Id	int64	Order identifier	Drop: ID
Order Item Cardprod Id	int64	Order-item product/card identifier	Drop: ID
Order Item Discount	float64	Discount amount on order item	Keep numeric
Order Item Discount Rate	float64	Discount rate on order item	Keep numeric
Order Item Id	int64	Order-item identifier	Drop: ID
Order Item Product Price	float64	Unit product price recorded for order item	Keep numeric
Order Item Profit Ratio	float64	Profit ratio for order item	Keep numeric
Order Item Quantity	int64	Quantity ordered	Keep numeric
Sales	float64	Gross sales amount	Keep numeric
Order Item Total	float64	Net/total value of order item	Keep numeric
Order Profit Per Order	float64	Profit per order	Keep numeric
Order Region	object	Order region	Keep; smoothed target encoding
Order State	object	Order destination state	Keep; smoothed target encoding
Order Status	object	Order processing/status category	Keep; one-hot encode
Order Zipcode	float64	Order postal/ZIP code	Drop: ~86% missing
Product Card Id	int64	Product/card identifier	Drop: ID
Product Category Id	int64	Product category identifier	Drop: ID
Product Description	float64	Product description field	Drop: 100% missing
Product Image	object	Product image URL/path	Drop: free-text/non-predictive
Product Name	object	Product name	Keep; smoothed target encoding
Product Price	float64	Listed product price	Keep numeric
Product Status	int64	Product status code	Drop: ID/constant in notebook
shipping date (DateOrders)	object	Shipment timestamp	Drop: post-order leakage
Shipping Mode	object	Selected shipping service/mode	Keep; one-hot encode